over here we can influence someone over there, but no matter how we do it, the procedure always involves someone or something traveling from here to there, and only when the someone or something gets there can the influence be exerted.

Physicists call this feature of the universe *locality*, emphasizing the point that you can directly affect only things that are next to you, that are local. Voodoo contravenes locality, since it involves doing something over here and affecting something over there without the need for anything to travel from here to there, but common experience leads us to think that verifiable, repeatable experiments would confirm locality.[3] And most do.

But a class of experiments performed during the last couple of decades has shown that something we do over here (such as measuring certain properties of a particle) *can* be subtly entwined with something that happens over there (such as the outcome of measuring certain properties of another distant particle), *without* anything being sent from here to there. While intuitively baffling, this phenomenon fully conforms to the laws of quantum mechanics, and was predicted using quantum mechanics long before the technology existed to do the experiment and observe, remarkably, that the prediction is correct. This sounds like voodoo; Einstein, who was among the first physicists to recognize—and sharply criticize—this possible feature of quantum mechanics, called it "spooky." But as we shall see, the long-distance links these experiments confirm are extremely delicate and are, in a precise sense, fundamentally beyond our ability to control.

Nevertheless, these results, coming from both theoretical and experimental considerations, strongly support the conclusion that the universe admits interconnections that are not local.[4] Something that happens over here can be entwined with something that happens over there even if nothing travels from here to there—and even if there isn't enough time for anything, even light, to travel between the events. This means that space cannot be thought of as it once was: intervening space, *regardless of how much there is*, does not ensure that two objects are separate, since quantum mechanics allows an entanglement, a kind of connection, to exist between them. A particle, like one of the countless number that make up you or me, can run but it can't hide. According to